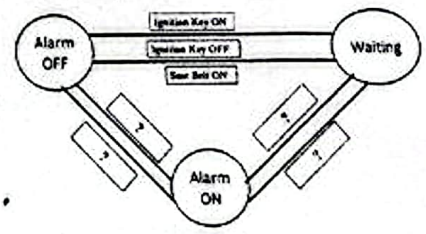
International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Spring-2025
Course Code: CSE-3523 **Course Title:** Microprocessors, Microcontrollers and Embedded Systems
Time: 2 hours 30 minutes **Total Marks: 50**

	Group-A	M	C
1a	Differentiate between maskable and non-maskable interrupts in 8086 with real-life examples.	3	2
b	Explain how a fixed interrupt vector can be generated using a DIP switch and 74ALS244 buffer during INTR in 8086. Include a neat and labeled circuit diagram.	4	2
c	Describe the sequence of operations performed by 8086 when a divide-by-zero operation occurs. Include the register changes and use of the stack during Type 0 interrupt.	3	2
	OR		
1. a	Suppose an external device generates an interrupt with vector number 60H. Describe how the CPU handles this interrupt after receiving the request. Including the steps involved in locating and executing the corresponding ISR.	5	2
b	Discuss the significance of the lower 3 interrupt types (Type 0–2) in 8086. Mention how each is triggered and handled.	5	2
2. Read the passage and answer the questions 2(a–c): Design a smart traffic signal system for a city that processes sensor inputs, prioritizes emergency vehicles, and reduces congestion using an embedded unit with multitasking, real-time interrupts, and optimized processing.			
a	Identify and justify 3 key processor features for the system. How does dual-core architecture improve multitasking?	3	
b	Based on this scenario, suggest which type of processor (e.g., embedded, dual-core, arithmetic, bit-slice) would best suit the system. Explain your choice with architectural and application justification.	4	
c	Explain how the use of cache memory and pipelining can enhance the performance of the traffic control system.	3	

	Group-B	
3. a	What are the differences between isolated I/O and memory-mapped I/O? Discuss their advantages and disadvantages with relevant instruction examples.	3
b	Describe the internal operation of a basic input interface using a tri-state buffer. How does the SEL (Select) signal help in transferring data from input switches to the CPU? Illustrate with a labeled diagram.	3
c	What is handshaking in microprocessor I/O systems? Explain Mode 1 input handshaking of 82C55 with STB, IBF, INTR, and RD signals, including a timing diagram.	4
4a	Differentiate microprocessor and microcontroller by architecture, functionality, and application, with one example each.	3
b	8051 microcontroller comes in a 40-pin DIP package. Draw and explain the pin diagram of the 8051, identifying the purpose of the following pins: 1.EA (External Access) 2. PSEN (Program Store Enable) 3.ALE (Address Latch Enable) 4.RST (Reset) 5. Port 0 and Port 3 special functions.	
c	Write an 8051 assembly language program to multiply two numbers stored at memory locations 40H and 41H. Store the result in registers A and B.	
	OR	
4a	Write a program to add the values stored at memory locations 40H and 41H, and store the result at 50H and 51H.	
b	Explain how a DC motor can be controlled using pin P1.3 and P1.4 of the 8051 microcontroller. Why is L293D necessary, and how do direction control signals work? Include circuit and code	
5. a	Look at the figure of FSM(Seat Belt Warning System) at right and answer questions (a-c): Identify all states and events in the FSM. Explain each state in the context of a car's seat belt warning system, complete missing transitions, and explain the FSM.	
b	Describe two conditions triggering transition from the 'Waiting' state and their effects. Provide another FSM example, e.g., a traffic light controller.	
c	Identify three limitations of traditional embedded design and their impact on performance. Explain co-design, its benefits, and give an RTOS-based example.	



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